divergence

a natural selection simulation. Rulebook

To start: open the hosted link, choose a Preset (or type a Seed), and press Start. Press Pause to freeze, 1 to 4 for speed. Click the World to inspect a Cell. In the side panel, read the Traits tab to see how a Trait changes over generations.

What this is

Divergence is a model of natural selection. Cells live in a World with Food, Toxin and varying Temperature. Cells that suit the World survive and reproduce, so their Traits become more common over generations. Over time the population can split into Species.

Controls

Drag to move the view. Scroll to zoom. Click a Cell to inspect it and see its sense range. Space pauses. Keys 1 to 4 change speed. F fits the World. T boosts the thermal map. Tab hides the charts panel. Escape cancels or closes.

Settings

The Seed fixes the World. Food abundance, Temperature range, zone scale, Toxin level and mutation rate shape it. Presets fill these in. The same Seed and settings always make the same starting World.

The World

Temperature varies smoothly from cold blue currents to hot coral zones. Food appears as glowing green pellets, more in some places than others. Toxin is a set of pink particles that drift, attract each other slightly and fade.

Traits

Each Cell inherits ten Traits: size, speed, sense range, preferred Temperature, Temperature tolerance, Toxin resistance, diet, reproduction threshold, sexual tendency and choosiness. Traits have benefits and trade-offs, so no Cell is best everywhere. Hover a Trait for its meaning, units and trade-off.

What the model simplifies

Species never merge once they have split. Cells live in an open World with a wall, not on a grid. Every Run is repeatable: the same Seed and settings start the same World, but later births involve chance, and Events you place change the outcome.

Reproduction

Below sexual tendency 0.50, a Cell divides at its energy threshold. At 0.50 or above, it needs a sexual partner within sense range with enough energy and mutual acceptance under both Cells' choosiness. Both pay energy; crossover takes each Trait and Brain weight from either parent, then mutation can make small random changes. Without a partner it waits; it never divides alone.

Species and colours

Cells begin white. A Cell's colour comes from its Genome, so similar Cells have similar colours. Grazers show a green inner core while high-diet hunters grow warm amber cores and front mandibles. A Species is a group of Cells whose Genomes are close together; a new Species appears when a group drifts far enough apart.

Charts

Traits shows one Trait over time or mean Generation, with a dashed line marking the Founders' starting average. Species shows the twenty Species with highest historical peaks; the rest form Other. Tree shows which Species split from which. Stats shows deaths in the latest ten minutes; legend totals cover the whole Run. Sexual share counts living Cells with sexual tendency at least 0.50, not births.

Events

Events change the World while a Run is going. Pick an Event from the bottom dock, set its size, then click the World to place it. Whole World applies without placement; for toxin spills, radius still sets quantity.

Save and exports

Save requests a Run download. Load restores a valid saved file. Export CSV requests two data files; your browser may ask permission for multiple downloads. Screenshot requests a PNG. Check your browser's downloads; the app cannot confirm files reached disk.

The five Events

Heat wave	Raises Temperature inside the circle, then lets it return to normal.
Ice age	Lowers Temperature inside the circle, then lets it return to normal.
Famine	Removes most Food inside the circle and slows new Food there for a while.
Food bloom	Adds extra Food inside the circle for a while.
Toxin spill	Releases Toxin inside the circle. It spreads slowly and fades.

Window size: at least 1280 by 720. Works in recent Chrome, Edge, Firefox and Safari.